STAT A411/611 Lecture Notes

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Office hours: MWF 12:00–12:45 **Class:** 2:30–3:45

Lecture Plan and Review

Today: Section 1.3.
Next Monday: Chapter 2.
Next Wednesday: Sections 3.1 and 3.2.
Last Wednesday: Sections 1.1, 1.2, and 1.3.

Review of Section 1.1

An experiment produces an outcome in a *sample space* S. Any subset of S is an *event*. An event may be described in words or written as a set. If event A occurs, its probability is denoted $P(A)$, where

$$0 \leq P(A) \leq 1.$$

Classical Formula (C.F.). If all outcomes in S are equally likely, then for any event A,

$$\boxed{P(A) = \frac{\#(A)}{\#(S)}}.$$

For a fair coin flipped twice,

$$S = \{HH, HT, TH, TT\}, \qquad P(\text{at least one head}) = \frac{3}{4}.$$

For a red die and a blue die, the eleven possible sums $2, 3, \ldots, 12$ are not equally likely. The 36 ordered pairs (R, B) are equally likely, and five of them give a sum of 8, so

$$P(\text{sum} = 8) = \frac{5}{36}.$$

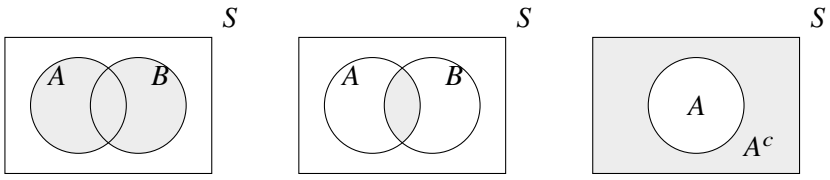
If one card is selected at random from a shuffled 52-card deck, there are 12 face cards (jack, queen, or king), giving

$$P(\text{face card}) = \frac{12}{52}.$$

Review of Section 1.2: Set Theory

For events A and B,

Union: $A \cup B = A$ OR B,
Intersection: $A \cap B = A$ AND B,
Complement: $A^c =$ NOT A.

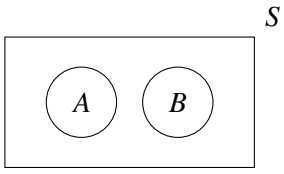


$A \cup B$ $A \cap B$ complement

Two events are *mutually exclusive* (disjoint) if they cannot occur at the same time:

$$A \cap B = \varnothing.$$

Equivalently, **at most one occurs**. More generally, $A_1, \ldots, A_n$ are mutually exclusive when $A_i \cap A_j = \varnothing$ for every $i \neq j$.



mutually exclusive events

Review of Section 1.3

I. Basic Rule of Counting (BRC). For k successive experiments or choices, if choice i has n_i possible outcomes, then the complete sequence has

$$\boxed{n_1 n_2 \cdots n_k}$$

possible different outcomes.

$$\underset{n_1}{E_1} \longrightarrow \underset{n_2}{E_2} \longrightarrow \underset{n_3}{E_3} \longrightarrow \cdots \longrightarrow \underset{n_k}{E_k}$$

Examples include menus, license plates, ZIP codes, passwords, and telephone numbers.

II. Permutations. Starting with n distinct elements, select k elements randomly and without replacement, where $1 \leq k \leq n$ and the order of selection matters. Each different ordered selection is a permutation. Using the Basic Rule of Counting,

$$\boxed{{}_nP_k \overset{\text{BRC}}{=} \overbrace{n}^{E_1}\,\overbrace{(n-1)}^{E_2}\cdots\overbrace{(n-k+1)}^{E_k} = \frac{n!}{(n-k)!}}.$$

The factorial notation used here includes

$$0! = 1, \qquad 5! = 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 = 120.$$

1 Section 1.3: Counting Rules, Continued

1.1 Permutations of All n Distinct Elements

The permutation formula is

$$ {}_nP_k = \frac{n!}{(n-k)!}. $$

For the special case $k = n$, all n distinct elements are selected and rearranged:

$$ e_1, e_2, \ldots, e_n \quad \longrightarrow \quad \text{an ordering of } e_1, e_2, \ldots, e_n. $$

Substituting $k = n$ gives the full chain

$$ {}_nP_n = \frac{n!}{(n-n)!} = \frac{n!}{0!} = \frac{n!}{1} = n!. $$

Thus there are $n!$ different orderings (rearrangements) of n distinct elements.
Example: FOREST. The six letters are all distinct. The different orderings begin and end schematically as

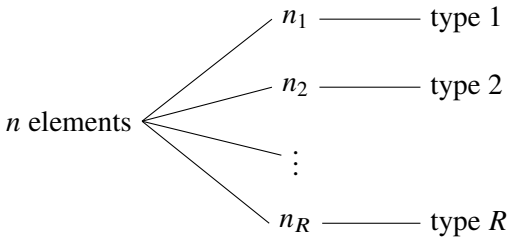
$$ \begin{array}{rl} 1. & \texttt{FOREST} \\ 2. & \texttt{FORETS} \\ & \vdots \\ 720. & \texttt{TSEROF} \end{array} \qquad \Longrightarrow \qquad 6! = 720. $$

1.2 III. Permutations of Indistinguishable Objects

Suppose n objects consist of R types. If type i occurs n_i times, then

$$ n_1 + n_2 + \cdots + n_R = n. $$

The structure can be pictured as



$$ n_1 + n_2 + \cdots + n_R = n $$

The number of different orderings is

$$ \boxed{\frac{n!}{n_1! n_2! \cdots n_R!}}. $$

For example, with 3 A's, 2 B's, 4 C's, and 1 D, $n = 10$ and the number of different orderings is

$$ \frac{10!}{3!\,2!\,4!\,1!}. $$

As a special case, if all n objects are distinct, then $n_1 = n_2 = \cdots = n_R = 1$ and

$$\frac{n!}{1!1!\cdots 1!} = \frac{n!}{1} = n!.$$

Word Examples

$$\begin{aligned}
\texttt{FOREST}: &\quad \frac{6!}{1!1!1!1!1!1!} = \frac{720}{1} = 6! = 720,\\
\texttt{PEPPER}: &\quad \frac{6!}{3!2!1!} = \frac{720}{6\cdot 2\cdot 1} = 60,\\
\texttt{ALASKA}: &\quad \frac{6!}{3!1!1!1!} = \frac{720}{6} = 120,\\
\texttt{IDAHO}: &\quad \frac{5!}{1!1!1!1!1!} = \frac{120}{1} = 120,\\
\texttt{MISSISSIPPI}: &\quad \frac{11!}{1!4!4!2!}.
\end{aligned}$$

1.3 IV. Combinations

Start with n distinct elements and select k elements randomly and without replacement, where $1 \le k \le n$. If the order of selection is *not* relevant, every possible selection is a *combination*.

For each combination of k selected elements, there are $k!$ different orderings (permutations). Therefore,

$${}_nP_k = k!\,{}_nC_k.$$

Solving for the number of combinations gives

$$\begin{aligned}
{}_nC_k &= \frac{{}_nP_k}{k!}\\
&= \frac{\dfrac{n!}{(n-k)!}}{k!}\\
&= \frac{n!}{k!(n-k)!}.
\end{aligned}$$

Thus

$$\boxed{{}_nC_k = \binom{n}{k} = \frac{n!}{k!(n-k)!}}.$$

The notation $\binom{n}{k}$ is read "n choose k."

Example: Division I Football

Suppose there are 132 Division I college football teams.

a) A ranked top 25 depends on order. For example, interchanging the first two teams gives a different ranking:

$$\begin{array}{c} 1.\ \text{OR} \\ 2.\ \text{OH St.} \\ \vdots \\ 25.\ \text{OK} \end{array} \qquad \begin{array}{c} 1.\ \text{OH St.} \\ 2.\ \text{OR} \\ \vdots \\ 25.\ \text{OK} \end{array} \qquad \text{(different rankings).}$$

Thus the number of possible rankings is

$$\boxed{{}_{132}P_{25} = \frac{132!}{107!} = 132 \cdot 131 \cdot 130 \cdots 108}.$$

b) Suppose the season ends with a 16-team playoff field. Only the set of teams matters, so order does not matter. A selection such as

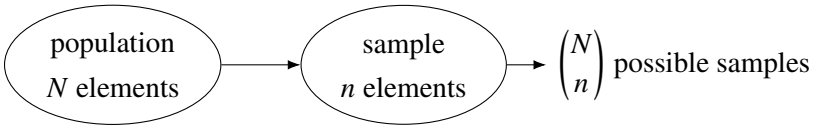
$$\{\text{OH St.}, \text{OR}, \ldots, \text{OK}\}$$

is the same selection regardless of how those 16 teams are listed. Therefore,

$$\boxed{{}_{132}C_{16} = \binom{132}{16} = \frac{132!}{116!16!}} = \frac{132 \cdot 131 \cdots 117}{16 \cdot 15 \cdots 1}.$$

Random Samples from a Population

In general, suppose a population contains N elements and a random sample contains n elements. Because the order of the selected sample does not matter, the question is: how many different samples are possible?



Therefore,

$$\boxed{\binom{N}{n} = \frac{N!}{n!(N-n)!}}.$$

Useful Combination Identities

From the combination formula,

$$\binom{n}{0} = \frac{n!}{0!\,n!} = 1, \qquad \binom{n}{n} = \frac{n!}{n!\,0!} = 1, \qquad \binom{n}{1} = \frac{n!}{1!(n-1)!} = n.$$

Also,

$$\begin{aligned} \binom{n}{n-k} &= \frac{n!}{(n-k)!\,[n-(n-k)]!} \\ &= \frac{n!}{(n-k)!\,k!} \\ &= \binom{n}{k}. \end{aligned}$$

Hence

$$\boxed{\binom{n}{k} = \binom{n}{n-k}}.$$

Examples include

$$\binom{6}{0} = 1, \qquad \binom{12}{12} = 1, \qquad \binom{12}{1} = 12, \qquad \binom{24}{23} = \binom{24}{1} = 24.$$

The number of five-card selections from a 52-card deck is

$$\binom{52}{5} = \frac{52!}{5!47!} = \frac{52 \cdot 51 \cdot 50 \cdot 49 \cdot 48}{5 \cdot 4 \cdot 3 \cdot 2 \cdot 1} = 13 \cdot 51 \cdot 10 \cdot 49 \cdot 8.$$

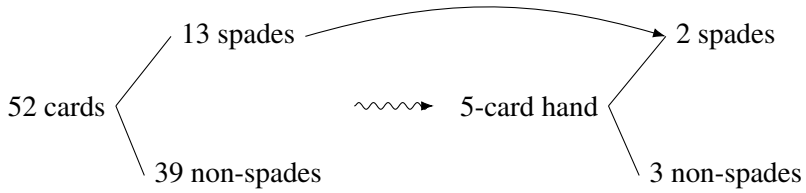
As a scale comparison, 10! = 3,628,800 > 3.5 million.

1.4 V. Probability Applications

Poker: Exactly Two Spades

A dealer shuffles a 52-card deck and gives the first player five randomly selected cards without replacement. What is the probability that exactly two of the five cards are spades?

There are 13 spades and 39 non-spades. An acceptable five-card hand contains two spades and three non-spades:



Because five-card hands are equally likely, the Classical Formula together with the BRC gives

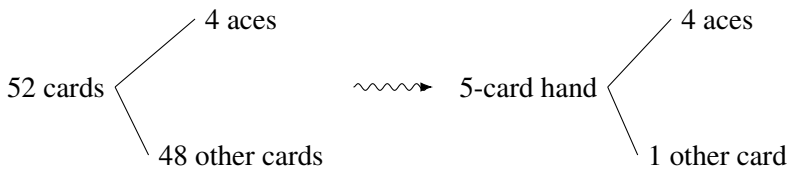
$$\boxed{P(\text{exactly 2 spades}) = \frac{\binom{13}{2}\binom{39}{3}}{\binom{52}{5}}}.$$

Expanding the combinations,

$$P(\text{exactly 2 spades}) = \frac{\left(\frac{13!}{2!11!}\right)\left(\frac{39!}{3!36!}\right)}{\frac{52!}{5!47!}} = \frac{\left(\frac{13 \cdot 12}{2 \cdot 1}\right)\left(\frac{39 \cdot 38 \cdot 37}{3 \cdot 2 \cdot 1}\right)}{\frac{52 \cdot 51 \cdot 50 \cdot 49 \cdot 48}{5 \cdot 4 \cdot 3 \cdot 2 \cdot 1}}.$$

Poker: All Four Aces

What is the probability that the first player gets all four aces? The population and five-card sample can be organized as



Thus

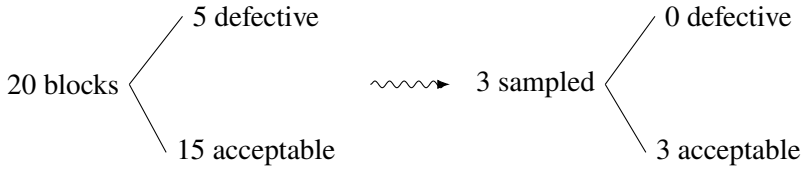
$$\boxed{P(\text{all 4 aces}) = \frac{\binom{4}{4}\binom{48}{1}}{\binom{52}{5}}}.$$

Expanding the combinations,

$$P(\text{all 4 aces}) = \frac{1 \cdot 48}{\frac{52 \cdot 51 \cdot 50 \cdot 49 \cdot 48}{5 \cdot 4 \cdot 3 \cdot 2 \cdot 1}} = \frac{120}{52 \cdot 51 \cdot 50 \cdot 49}.$$

Example 1.3.6: Lot Acceptance

A lot contains 20 blocks: 5 defective and 15 acceptable. Three blocks are sampled. The entire lot is accepted when all three sampled blocks are acceptable.



Therefore,

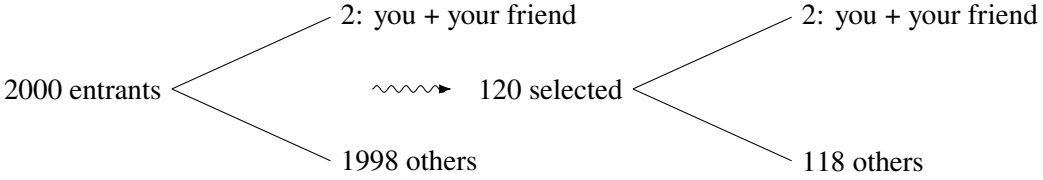
$$\boxed{P(\text{lot accepted}) = P(\text{all 3 sampled blocks are acceptable}) = \frac{\binom{5}{0}\binom{15}{3}}{\binom{20}{3}}}.$$

Expanding,

$$P(\text{lot accepted}) = \frac{1 \cdot \frac{15 \cdot 14 \cdot 13}{3 \cdot 2 \cdot 1}}{\frac{20 \cdot 19 \cdot 18}{3 \cdot 2 \cdot 1}} = \frac{15 \cdot 14 \cdot 13}{20 \cdot 19 \cdot 18}.$$

Additional Worked Problems

Problem 20(c). This problem models the McNeil River brown-bear viewing permit lottery in Alaska. From 2000 entrants, a group of 120 is selected. Assume you and a friend are both entered. The setup is



Therefore,

$$\boxed{P(\text{you and your friend are both selected}) = \frac{\binom{2}{2}\binom{1998}{118}}{\binom{2000}{120}}}.$$

Expanding and canceling,

$$\begin{aligned} P &= \frac{1 \cdot \dfrac{1998!}{118!\,1880!}}{\dfrac{2000!}{120!\,1880!}} \\ &= \frac{1998!}{2000!} \cdot \frac{120!}{118!} \\ &= \frac{1}{2000 \cdot 1999}\,(120 \cdot 119) \\ &= \frac{120 \cdot 119}{2000 \cdot 1999}. \end{aligned}$$

Problem 32. A computer system uses passwords consisting of five letters L_1, L_2, L_3, L_4, L_5 followed by one digit D_1.

a) With unrestricted choices,

$$26 \cdot 26 \cdot 26 \cdot 26 \cdot 26 \cdot 10 = 26^5 \cdot 10.$$

b) Suppose the password contains exactly three A's, two B's, and ends in an even digit. The five letters can be arranged in

$$\frac{5!}{3!2!} = 10$$

ways. The arrangements can be shown schematically as

1. `AAABB`
2. `AABAB`

$\vdots$

10. `BBAAA`

and the final digit has five possible even values $\{0, 2, 4, 6, 8\}$. Thus there are

$$10 \cdot 5 = 50$$

possible passwords satisfying the remembered pattern.

A further repeated-symbol example uses 3 A's, 2 B's, 1 C, 4 D's, and 2 E's. The number of different arrangements is

$$\frac{12!}{3!\,2!\,1!\,4!\,2!}.$$

Returning to Problem 32, if the password has been forgotten but the pattern above is remembered, a valid guess such as `ABABA6` is one of 50 possibilities. Therefore, the probability of guessing the password correctly on the first attempt is

$$\frac{1}{50} = 0.02 = 2\%.$$

Homework and Next Class

Homework: 2, 4, 6, 8, 10, 12, 14, 18, 20, 22, and 26.
Review: all except 38.

Next Monday: Chapter 2.
Next Wednesday: Sections 3.1 and 3.2.